

The default Debian 9 installation does not have the *sudo* package installed. Log in as the root user, and install the *sudo* package on all the three VMs. The ‘debian’ user also requires *sudo* access:

```
root@debian:~# apt-get install sudo

root@debian:~# adduser debian sudo
Adding user `debian' to group `sudo'...
Adding user debian to group sudo
Done.
```

You can now test connectivity from Ansible to the individual Consul nodes as well as collectively, by using the following commands:

```
$ ansible -i inventory/kvm/inventory consul1 -m ping
consul1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}

$ ansible -i inventory/kvm/inventory consul2 -m ping
consul2 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}

$ ansible -i inventory/kvm/inventory consul3 -m ping
consul3 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}

$ ansible -i inventory/kvm/inventory all -m ping
consul2 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
consul3 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
consul1 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

Consul

The first step is to install Consul on all the nodes. The software package repository is updated, and a few network tools are installed. The ‘consul’ binary is extracted from a Zip file obtained at HashiCorp’s website and copied to */usr/local/bin/consul*. The execution of the binary is

verified. A ‘consul’ user account is created in the system, and the */var/consul* directory is created and owned by the ‘consul’ user. The playbook to install Consul is provided below for reference:

```
---
- name: Install Consul
  hosts: all
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [consul]

tasks:
  - name: Update the software package repository
    apt:
      update_cache: yes

  - name: Install dependencies
    package:
      name: "{{ item }}"
      state: latest
    with_items:
      - curl
      - net-tools
      - unzip

  - name: Install consul
    unarchive:
      src: https://releases.hashicorp.com/consul/1.1.0/consul_1.1.0_linux_amd64.zip
      dest: /usr/local/bin
      remote_src: yes

  - name: Verify consul installation
    shell: "consul --version"
    register: consul_version

  - assert:
      that:
        - "'Consul' in consul_version.stdout"

  - name: Create consul user
    user:
      name: consul

  - name: Create Consul /var directory
    file:
      path: /var/consul
      state: directory
      owner: consul
      group: consul
      mode: 0755
```

The above playbook can be invoked using the